MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, RECEIVING

## TYPE 5845

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

**DESCRIPTION:** Double diode, miniature, temperature limited

| | | |
|---|---|---|
| Outline | --- | 5-2(EIA) |
| Base | --- | E7-1 |
| Envelope | --- | T5-1/2 |
| Cathode | --- | Tungsten filament |

Base connections:

| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| Element | --- | 2a | nc | f | f | 1a | nc | fct |

**ABSOLUTE-MAXIMUM RATINGS:**

| Parameter: | Ef | Ebb | Ib (Note 1) | Rl | Orientation | TE | Alt |
|---|---|---|---|---|---|---|---|
| Unit: | Vac | Vdc | mAdc | Meg | --- | °C | ft |
| Maximum: | 4.0 | 275 | 1.1 | --- | Note 2 | 50 | Note 3 |
| Minimum: | --- | --- | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | Adj | 250 | 0.05 | 1.0 | --- | --- | --- |

**GENERAL:**

Qualification - Required

Ⓐ denotes changes



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | | | | | MIN | MAX | |
| | Quality conformance inspection, part 1 | | | | | | | |
| 1261 | Filament voltage | See notes 1 and 4 | 0.65 | II | Ef | 2.6 | 3.4 | Vac |
| --- | Leakage anode current | Ef = 0 (see notes 1 and 5) | 0.65 | II | Ib | --- | 5 | μAdc |
| --- | Stability | See notes 1, 4, and 6 | 0.65 | II | Ib | 40 | 60 | μAdc |
| 1201 | Short and discontinuity detection | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1301 | Heater current | Ef = 3.0 Vac (see note 4) | 4.0 | S3 | If | 340 | 390 | mAac |
| 1256 | Electrode current (1) (anode) | Ef = 3.0 Vac (see notes 1 and 4) | --- | --- | Ib | 10 | 80 | μAdc |
| 1256 | Electrode current (2) (anode) | Ef = 3.25 Vac (see notes 1 and 4) | --- | --- | Ib | 25 | 275 | μAdc |
| 1031 | Vibration fatigue | F = 60 Hz; 22G (see notes 1, 4, and 7) | --- | --- | --- | --- | --- | --- |
| --- | Vibration stability | F = 60 Hz; 22G (see notes 1, 4, and 8) | --- | --- | Ib | 40 | 60 | μAdc |
| 1041 | Shock (1) | No voltages; 500G (see note 9) | --- | --- | --- | --- | --- | --- |
| 1041 | Shock (2) | 200G (see notes 1, 4, and 10) | --- | --- | --- | --- | --- | --- |
| (A) --- | Filament short | See notes 1 and 11) | --- | --- | --- | --- | --- | --- |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| 1501 | Intermittent life test | Group A; Ef = 1.5 Vac (see notes 1 and 12) | --- | --- | --- | --- | --- | --- |
| --- | Intermittent life-test end point (500 hours) | Filament voltage | --- | --- | Ef | --- | 3.8 | Vac |

NOTES:

1. Tie 1a to 2a.
2. During operation the tube shall be mounted either in a vertical position, base up or down, or in a horizontal position with the plane containing base pins 3 and 4 vertical.
3. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
4. Connect filament sections in series.



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NOTES: -Continued

5. Test duration shall be 2-seconds minimum. Tubes with leakage not exceeding 10 μA are acceptable providing the increase of leakage after the first 2-seconds does not exceed 5 μA over a 1-minute period.

6. Adjust Ef for Ib = 50 μAdc. Operate for 3 seconds and, if necessary, readjust Ef for Ib = 50 μAdc. Operate for 5 additional seconds without additional adjustment. At the end of the 5-second period, Ib shall meet the specified limits.

7. All general test conditions shall be applied. The tube under test shall be subjected to simple harmonic vibration of the specified frequency and peak acceleration for a period of 5 minutes in each of the three positions X, Y, and Z. Filament breakage shall be the only criterion for rejection.

8. All general test conditions shall be applied. The tube under test shall be subjected to simple harmonic vibration of the specified frequency and peak acceleration for a period of 10 seconds in the X position. After vibration, anode current shall be within the specified limits.

9. Each tube shall be subjected to one hammer blow in each of the four positions.

10. All general test conditions shall be applied to the tube under test during shock (2) test. Each tube shall be subjected to one hammer blow. This blow shall be in position X.

11. Electrical rejects may be used for this test. Tubes shall be tested for filament to anode shorting under ultimate filament opening (burnout) conditions. When filament is opened, a short shall occur between filament and anode. If rejection of a lot is indicated by failure to pass the sampling plan, the remaining tubes in a lot may be visually inspected for filament tension and anode shorting bar position and those tubes deemed capable of passing this test may be formed into a new lot. The newly formed lot shall then be resampled for this test. Failure to pass the retest shall be cause for rejection of the newly formed lot.

12. Connect filament sections in parallel.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 17
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2515)



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